
RE: Request for technical information (Universidad Nacional de Rafaela)

Desde Blake Hasty <Blake.Hasty@ametek.com>

Fecha Jue 24/10/31 9:00 AM

Para Alphasense Customer Service <sensors.alphasense@ametek.com>; Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>

Gustavo,

The relation to humidity very closely follows ideal gas law which can be adjusted for. This would offset the molecule size variability that is a component of the differences in humidity. The pressure though is a completely different function. Our sensors need to see roughly a 0.5L /min of flow and they are calibrated based on that specific flow rate. In the even that you apply greater pressure directly into the sensor, the resonance time internal with the electrolyte will not allow a complete reaction to happen and therefore give an erroneous measurement. This is mitigated to a point when flow is across the face (instead of directly into to face) of the sensors with the orifice opening which can control exposure should the gas stream above be greater than 0.5L/min. This will be the case for any and all electrochemical gas sensors. And depending on your measurement range needs, the more you control the environmental conditions and flow stream, the more reliable and accurate data you will receive. We are just a small piece of that puzzle. The rest is on the device that you are looking to integrate us into. Hope this helps.

Blake Hasty

OEM Sales Manager – USA/Americas



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From: Alphasense Customer Service <sensors.alphasense@ametek.com>

Sent: Thursday, October 31, 2024 3:26 AM

To: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>

Cc: Blake Hasty <Blake.Hasty@ametek.com>

Subject: RE: Request for technical information (Universidad Nacional de Rafaela)

Hi Gustavo,

Thank you very much for your email.

You would have to make some experiments with humidity.

Best would be to exposed the sensor to different humidity levels and see how that might change the readings.

Same with the pressure, but not sure how exactly to do it.

We have not performed any testing like that, thus not sure how to advice.

Krzysztof

Alphasense Customer Service



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From: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Sent: 30 October 2024 18:36
To: Alphasense Customer Service <sensors.alphasense@ametek.com>
Subject: RE: Request for technical information (Universidad Nacional de Rafaela)

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Hi Krzysztof

Thank you very much for your reply.

According to what you tell me, the sensor measurement is a function of the humidity and the pressure with which the air circulates in front of the sensor, is this correct?

Regarding the temperature variations, you sent me an application note so that we can be independent of it if we have the capacity to measure the difference at the reference temperature, in relation to humidity and pressure, is there any way to be independent and calculate the real PPB value of NO? or otherwise the sensor would not be useful for the measurement that we want to make of NO in the expired air of humans?

Sincerely,

Gustavo

De: Alphasense Customer Service <sensors.alphasense@ametek.com>
Enviado: miércoles, octubre 30, 2024 7:09 AM
Para: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Asunto: RE: Request for technical information (Universidad Nacional de Rafaela)

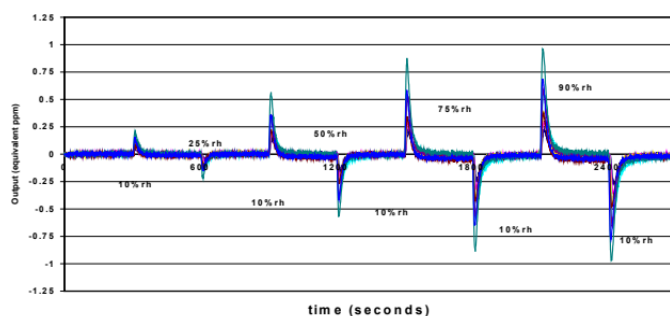
Hello Gustavo,

Thank you very much for your email.

NO B4 offers the lowest detection level from all our NO cells.

Please not when you blow on the sensor you are creating a humidity change, and pressure change to which sensor will respond.

This impact can be either positive – will add to the signal, or negative – you will see decrease in concentration.



Temperature change will also have an impact.

That are the factors you would have to consider during the research.

Unfortunately, we have not performed any testing similar to your application.

Not much help, sorry. Let me know if you will have any other questions.

Best regards

Krzysztof

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From: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Sent: 29 October 2024 17:55
To: Alphasense Customer Service <sensors.alphasense@ametek.com>
Subject: RE: Request for technical information (Universidad Nacional de Rafaela)

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Hi Krzysztof,

We would love to consult you again about the development we are developing in the University laboratory with NO-B2, we are trying to develop a prototype of a device to measure exhaled nitric oxide (usually called FEno) and currently we measure the outputs of the ISB with a 0.15mV step ADC and when testing the sensor we see the following: after a while the outputs of the sensor in contact with the air are:
WE= 272.9mV AE=248.8mV (which is close to the offset given in the ISB data sheet)

The second test is to blow through the hose to one of the inputs to the ISB clamping kit, and the values change to the following:
WE= 165.3mV AE=254.2mV

According to the ISB data sheet with a sensitivity of 432mV/ppm, and taking the variation of WE is equal to 107mv, this would be equivalent to a measurement of 247ppb of NO in the air exhaled by me... which is not possible because it is much higher than the NO that a human being can exhale, this raises a doubt as to whether we are doing something wrong.

On the other hand... in the sensor data sheet... it indicates that the output voltage should rise and not fall when detecting NO, it is for these reasons that we are contacting you.

Figure 3 Response to 200ppb NO

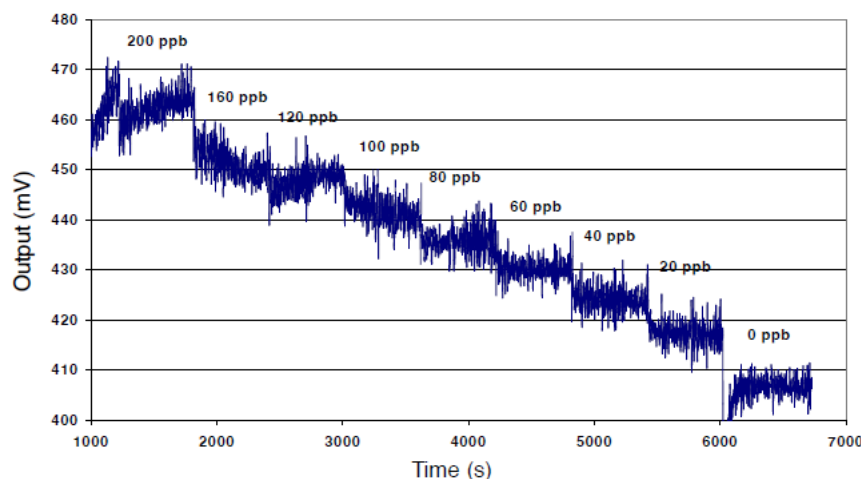


Figure 3 shows response to 200ppb NC
Use of Alphasense ISB circuit reduces n
with the opportunity of digital smooting
noise even further.

Are we using the correct sensor and in the correct way?

Best regards,

Gustavo.-

De: Alphasense Customer Service <sensors.alphasense@ametek.com>
Enviado: martes, octubre 08, 2024 1:21 PM
Para: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Asunto: RE: Request for technical information (Universidad Nacional de Rafaela)

Hi Gustavo,

Please do not remove it. It is dust filter which protects working electrode below it.

Best regards

Krzysztof

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From: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Sent: 08 October 2024 14:57
To: Alphasense Customer Service <sensors.alphasense@ametek.com>
Subject: RE: Request for technical information (Universidad Nacional de Rafaela)

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Hi Krzysztof,

Thank you very much for your mail.

One more question please, should the white sticker on the sensor be removed before using it or is it some kind of filter?

Best regards,

Gustavo.-

De: Alphasense Customer Service <sensors.alphasense@ametek.com>
Enviado: martes, octubre 08, 2024 4:00 AM
Para: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Asunto: RE: Request for technical information (Universidad Nacional de Rafaela)

Hi Gustavo,

I hope you are doing well.

Please leave it under power for a day and readings will go down.

NO requires Bias voltage, thus best is to keep them continuously under power – for best signal stability.

You will see signal going down with time.

Let me know if you will have any other questions.

Best regards

Krzysztof

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From: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Sent: 07 October 2024 23:06
To: Alphasense Customer Service <sensors.alphasense@ametek.com>
Subject: RE: Request for technical information (Universidad Nacional de Rafaela)

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Hi Krzysztof,

Some time ago we received the NO-B4 sensor and the corresponding ISB board already assembled from your factory, and today I was able to start testing.

My first test was to power the ISB board and measure the OP1 and OP2 outputs before connecting them to the ADS115 ADC inputs, but I was surprised to find that the outputs did not present the values that the ISB user manual theoretically gave; instead of delivering values of around 270 mV, the measurements were OP1 = 4870mV, and OP2 = 4933mV (with $V_{in} = 5071\text{mV}$). We still have the sensor with the sticker cover that came from the factory.

Do you have any idea why the OP1 and OP2 values are so high?

Thank you very much in advance, best regards

Gustavo.-

De: Alphasense Customer Service <sensors.alphasense@ametek.com>

Enviado: martes, marzo 26, 2024 5:03 AM

Para: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>

Asunto: RE: Request for technical information (Universidad Nacional de Rafaela)

Hello Gustavo,

Thank you very much for your order.

Placing An Order

To place an order with us, please send your order to orders.alphasense@ametek.com. Please provide the following details if they are not included on your purchase order

- Company Name
- Delivery address along with contact name and details
- Invoice address along with contact name and details if different from the above
- Purchase Order number
- VAT number (where applicable)
- Your preferred payment method (Credit Card or Bank Transfer)
- If using your preferred freight forwarder - please provide details of provider and account number

Please note – we are unable to accept payments via AMEX or PayPal.

Any enquiries regarding quantities please contact my colleagues at orders. I will be happy to assist with quotes and tech support.

Calibration – expose the sensor to known gas concentration and adjust the readings in the software.

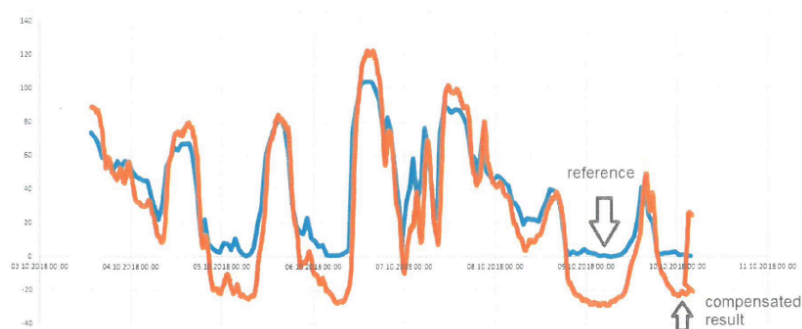
Attached app note might be useful with mV to ppb conversion.

With a four electrode sensors, the Working (WE1) and Auxiliary (WE2; AUX) electrode baseline are very similar. As ambient conditions are the same for both, a baseline shift on the Working electrode due to a temperature change will be accompanied by a shift on the Auxiliary electrode. When a Working electrode is exposed to a target gas it produces an additional signal proportional to the gas concentration, at the same time signal from Auxiliary electrode is unaffected. Subtracting two signals will give approximately a baseline compensated output due entirely to exposure to gas. We are still working how to exactly compensate using the auxiliary, it is different for each sensor type.

Algorithms are provided as suggestions for attempting to correct for temperature changes only and should not be regarded as a complete solution for providing gas concentration results that meet and accuracy acceptable for your application.

For the calibration of a low cost air quality network, one strategy would be to initially cluster your sensor nodes together around a reference station. Preferably they should be about 1m of the station to ensure as best as possible, the same gas sample is within this shared environment volume. What flows will then be a process of sensor calibrations to your environment of your sensors. After applying your chosen algorithms, you may find that the results do not meet expectations for accuracy when compared to reference values. You may find the gas concentrations are negative compared to reference values or appear to be much smaller or larger than reference values but in general the sensor output is following the reference concentration. Respectively these are offset errors (which can also be in the positive direction) and gain errors. They can be corrected by simply applying the correction factors.

For example:



As you can see above, there is a good correlation between the reference station and our cell. By applying a simple correction factor you can match our readings with the reference.

Please let us know if you will have any questions.

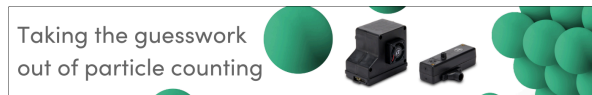
Best regards

Krzysztof

Alphasense Customer Service



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From: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Sent: Monday, March 25, 2024 11:08 PM
To: Alphasense Customer Service <sensors.alphasense@ametek.com>
Subject: Re: Request for technical information (Universidad Nacional de Rafaela)

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Dear Krzysztof,

Thank you very much for your prompt response.

Regarding the pro forma, we would like to increase the number of NO-B4 sensors to 2 (two). From what I understand, we follow everything related to payment and shipping through the email orders.alphasense@ametek.com, is that correct?

In relation to the sensor calibration every a certain amount of time, I understand that it could be done by hardware in a circuit external to the ISB on the signal of its output, or also by software using small gas bottles with ppb of NO already informed by the manufacturer, could it be correct?

With respect to the ISB, are the outputs OP1 and OP2 differential and equal? or should OP2-OP1 be performed to get the output shown in the ISB brochure?

Many Thanks, Sincerely,

Ing. Gustavo Castro
Universidad Nacional de Rafaela
gustavo.castro@unraf.edu.ar
+54 9 3492 57-1000

El mié, 13 mar 2024 a la(s) 6:02 a.m., Alphasense Customer Service (sensors.alphasense@ametek.com) escribió:

Morning,

Thank you very much for your email.

A4 / B4 are 4 electrode sensors in which 4th electrode is a correction for Zero Current and Temperature Zero Current shift.

With a four electrode sensors, the Working (WE1) and Auxiliary (WE2; AUX) electrode baseline are very similar. As ambient conditions are the same for both, a baseline shift on the Working electrode due to a temperature change will be accompanied by a shift on the Auxiliary electrode. When a Working electrode is exposed to a target gas it produces an additional signal proportional to the gas concentration, at the same time signal from Auxiliary electrode is unaffected. Subtracting two signals will give approximately a baseline compensated output due entirely to exposure to gas. We are still working how to exactly compensate using the auxiliary, it is different for each sensor type.

A4 small size (20mm diameter) but with good sensitivity and reproducibility. Designed for mobile air quality networks where space is at a premium.

B4 larger size (32mm diameter) same technology as the A4 but more robust when exposed to outdoor environments 24/7/ Designed for fixed site air quality networks where long term reliability is critical.

NO-B4 [alphasense_no-b4_datasheet_en_1.pdf\(azureedge.net\)](#).

As both types require low noise potentiostat circuit for best sensitivity and resolution we have introduced low noise interface boards.

AFE - for A4 sensors; between 2 and 4 cells on one board [alphasense_afe_datasheet_en_1.pdf\(azureedge.net\)](#).

ISB - Individual Sensor Boards for B4 sensors [alphasense_isb_datasheet_en_1.pdf \(azureedge.net\)](#)

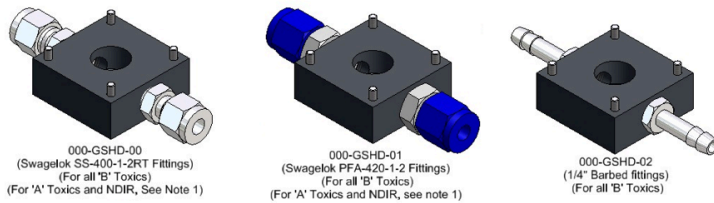
Current consumption <800uA working at the full Board scale (per channel). Please note that they do not contain the shorting FET so stabilisation time (around 2 hours) is needed if the power source would be removed from the circuit. (they have been designed for continuous use)

Re-calibration schedule will depend on the environment sensors will work in. More frequent for the sensors working in a heavy environment comparing to let say clean lab. Second parameter to consider is the accuracy and the detection level that you planing to achieve, thus high accuracy will require more frequent calibration.

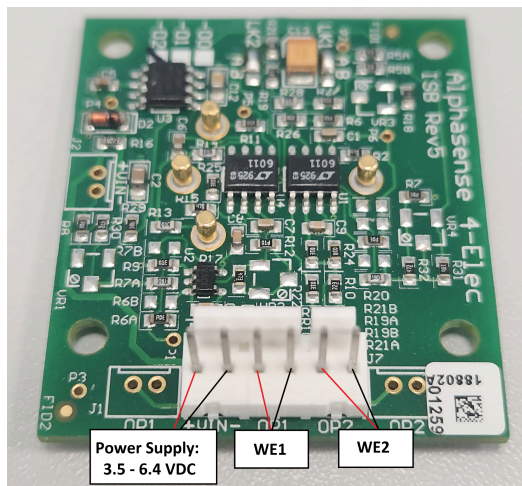
With all the sensors (A4 or B4) we will provide you with our initial calibration, thus you would be able to use them immediately.

NO sensors require Bias voltage (+200mV) to be fully operational. That is why they need to be continuously under power for best signal stability.

Available gas hoods:



ISB connection:



Attached our quote as requested.

Please have a look and let us know if you will have any questions.

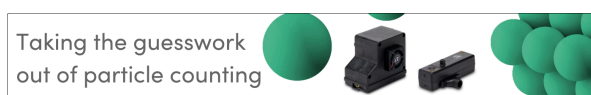
Best regards

Krzysztof Janiuk
Sales and Technical Support Manager

Alphasense Customer Service



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From: Gustavo Pablo Castro Abdallah <gustavo.castro@unraf.edu.ar>
Sent: Tuesday, March 12, 2024 5:55 PM
To: Alphasense Customer Service <sensors.alphasense@ametek.com>
Subject: Request for technical information

NOTICE This came from an external source. Use caution when replying, clicking links, or opening attachments.

Dear Sir or Madam,

my name is Gustavo Castro and I work in the Digital Transformation laboratory of the National University of Rafaela, Argentina; We will contact you to please request information (in addition to that published on your website) about the NO - B4 nitric oxide sensor and its individual sensor board. (ISB) 000-0ISB-01. In the laboratory we are developing research work on the prevention of respiratory diseases in children associated with the theory of its relationship with the nitric oxide exhaled by infants for which we have chosen to use their products, so it is vitally important to have with all the extra technical information about their products, as well as related application notes and quality certificates that they hold and to include in our research.

We would also appreciate information on where both parts can be purchased, as well as the optional ISB accessories.

I look forward to hearing from you, Sincerely,

Ing. Gustavo Castro
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